



Sir Syed University of Engineering & Technology

END SEMESTER EXAMINATIONS FALL 2021

Faculty: FoCAS Department Computer Science & IT

Semester 4th Degree Program BSCS Batch 2020S

Course Code: MS-208 Course Title: Multivariate Calculus Credit Hours: 3

Timings: 3 Hours

Date of Examination: 12-02-22

Max Marks: 50

Instructions:

- Please don't write anything on the Question Paper except for your Roll No. and Name.
- The paper must be solved / attempted on Printed Answer books provided by the Examinations Department.

Roll No.	<u>214</u>	Name	<u>Wahid</u>	A' Copy Serial No.	
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Q.No.1

10 Marks

a) Verify Stoke's theorem for the vector field $\mathbf{F} = (2x - y)\mathbf{i} - yz^2\mathbf{j} - y^2z\mathbf{k}$ over the upper half surface of $x^2 + y^2 + z^2 = 1$ and C is its boundary.

b) (i) Find $\int_1^3 \int_{-1}^1 \int_{-2}^4 3(x^2y + y^2z) dz dy dx$

(ii) Transform the equation $x^2 + y^2 + z^2 = 10$ into spherical Coordinates.

Q.No.2

10 Marks

a) By finding Fourier Series for the function $f(x) = x^2/4$ ($-\pi < x < \pi$)

show that $\frac{\pi^2}{6} = 1 + \frac{1}{4} + \frac{1}{9} + \frac{1}{16} + \dots$

b) Find $\frac{dw}{dt}$ using chain rule: $w = \ln(x + y)$; $x = e^{-2t}$, $y = t^3 - t^2$.

Q.No.3

10 Marks

a) Find the Fourier series of the following

$$f(x) = \begin{cases} x & \text{if } -\pi < x < 0 \\ \pi - x & \text{if } 0 < x < \pi \end{cases}$$



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b) Find the Fourier cosine series.

$$f(x) = e^x \quad (0 < x < p)$$

Q.No.4

10 Marks

a) Verify Green's theorem in the plane for $\oint (xy + y^2)dx + x^2 dy$ where c is the closed curve of the region bounded by $x=y$ and $x^2=y$.

b) Evaluate $\iint_D 2e^{x^2+y^2} dA$, D is the portion of the region between the circles $x^2 + y^2 = 25$ and $x^2 + y^2 = 49$ that lies in the first quadrant.

Q.No.5

10 Marks

a) Evaluate $\iint_S F \cdot n \, dS$ where $F = 4xz\mathbf{i} - y^2\mathbf{j} + yz\mathbf{k}$ and S is the surface of the cube bounded by $(0,0,0)$ to $(1,1,1)$

b) Prove that $\nabla^2 \left(\frac{1}{r} \right) = 0$



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Faculty: Faculty of Computing & Applied Sciences Department: Computer Science & Information Technology

Semester 4th Degree Program BS(CS) Batch 2020S

Course Code: CS-222T Course Title: Introduction to Operating System Credit Hours: 3+0

Timings: 3 Hours Date of Examination: 09-02-2022 Max Marks: 50

Instructions:

- Please don't write anything on the Question Paper except for your Roll No. and Name.
- The paper must be solved / attempted on Printed Answer books provided by the Examinations Department.
- Attempt all questions.

Roll No.	Name	'A' Copy Serial No.
214		

Q.No.1 (5+5 Marks)

- (a) Consider a logical address space of 8 pages of 1024 words each, mapped onto a physical memory of 32 frames. How many bits are there in the physical address and logical address respectively?
- (b) What is the effect of allowing two entries in a page table to point to the same page frame in memory? Explain how this effect could be used to decrease the amount of time needed to copy a large amount of memory from one place to another. What effect would updating some byte on the one page have on the other page?

Q.No.2 (5+5 Marks)

- (a) We have an operating system for a machine that uses base and limit registers, but we have modified the machine to provide a page table. Can the page tables be set up to simulate base and limit registers? Why can they be, or why can they not be?
- (b) Consider the following page reference string: 1, 2, 3, 4, 5, 1, 3, 1, 6, 3, 2, 3. Assuming frame size 4, how many page faults would occur for the following page replacement algorithms? FIFO, LRU, Optimal page replacement algorithm.



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Q.No.3

(5+5 Marks)

- (a) What is the meaning of the term busy waiting? What other kinds of waiting are there in an operating system? Can busy waiting be avoided altogether? Explain your answer.
- (b) Describe the basic idea of the Bakery algorithm using your own words. You should mention which problem the algorithm is trying to solve, and how it is solving it (conceptually). There is no need for pseudo code.

Q.No.4

(5+5 Marks)

- (a) Demonstrate the working of Peterson's solution for process synchronization with example.
- (b) A distributed system has two sites, A and B. Consider whether site A can distinguish among the following:
 - 1. B goes down.
 - 2. The link between A and B goes down.
 - 3. B is extremely overloaded, and its response time is 100 times longer than normal.What implications does your answer have for recovery in distributed systems?

Q.No.5

(5+5 Marks)

- (a) Consider a system with 4 processes P1, P2, P3, and P4. Arrival time of processes is 0,1,2,3 with burst time requirements of these processes are 4, 1, 8, and 1 respectively. Then what is the completion time of Process P1. Assuming Round Robin Scheduling with Time Quantum is 1.
- (b) Consider 6 processes P1,P2,P3,P4,P5,P6 with arrival time 6,3,4,1,2,5 and burst time 1,3,6,5,2,1 respectively. If the CPU scheduling policy is SJF non-preemptive, calculate the average waiting time and average turnaround time.



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END SEMESTER EXAMINATIONS FALL 2021

Faculty: Computing & Applied Sciences Department Computer Science & IT

Semester IV Degree Program BSCS Batch 2020-S

Course Code: CS-214 Course Title: Theory of Automata & Formal Languages Credit Hours: 3+0

Timings: 3 Hours

Date of Examination: 01-02-2022

Max Marks: 50

Instructions:

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Q.No.1

8 Marks

Make the CFG for the following and draw the parse tree for one valid word of each

a> All words that have at least three character length but having **ba** as a starting sub-string

b> $d^m c^x a^n b^n e^m$ where $m>0$; and $n,x\geq 0$.

c> $a^x b^{n+m} d^y e^{n+m}$ where $x,y>0$; and $n,m\geq 0$.

d> $b^{n+2} c^{n+3}$ where $n\geq 0$.

Q.No.2

4 Marks

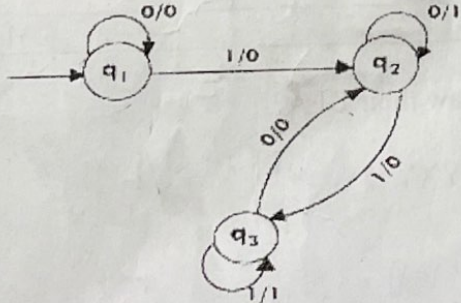
Make a Moore Machine that count the occurrence of pattern **aba** and **bab** and **bba**.

Q.No.3

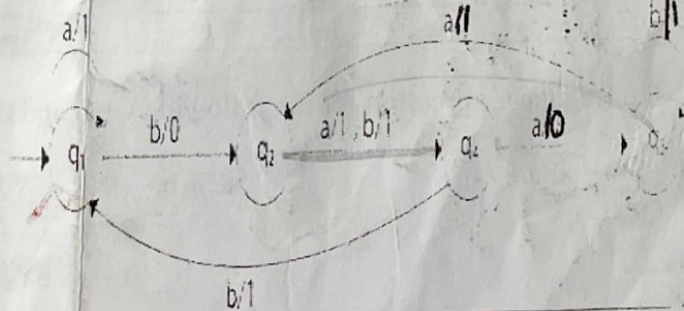
4 Marks

Convert the given mealy machines into Moore machine?

a>



b>





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Q.No.4

9 Marks

Convert the given Context Free Grammar into Chomsky's Normal Form

a> $S \rightarrow AaBb$

$A \rightarrow abA \mid ba$

$B \rightarrow bBa \mid \epsilon$

b> $S \rightarrow XabcY$

$X \rightarrow Y \mid ca \mid cY$

$Y \rightarrow bY \mid \epsilon$

c> $S \rightarrow ABC$

$A \rightarrow aaA \mid baB \mid \epsilon$

$B \rightarrow bbB \mid C \mid \epsilon$

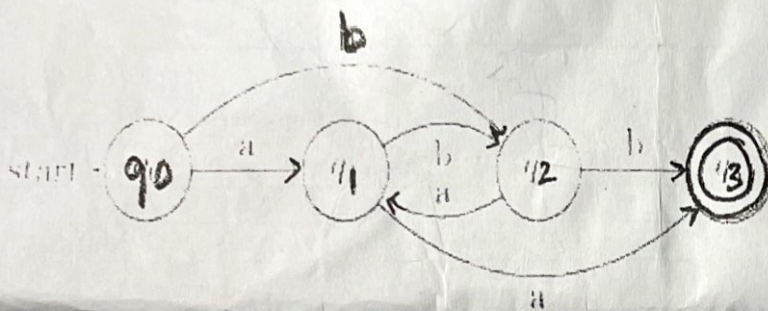
$C \rightarrow \epsilon$

Q.No.5

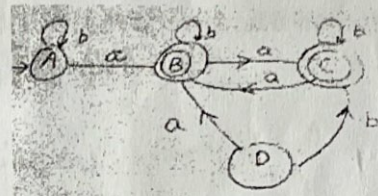
9 Marks

Write the Regular Expression for the following with complete steps.

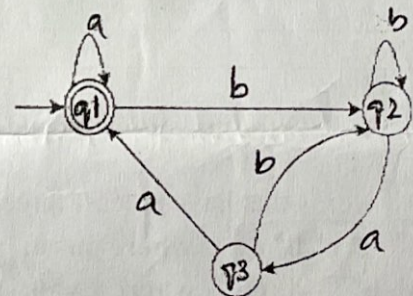
a>



b>



c>



Q.No.6

12 Marks

Draw the Push Down Automata for the following.

a> $e^x a^n b^m c^{m+n} d^y \quad |n > 0, m, x, y \geq 0$

b> $a^m b^n c^x d^n e^m \quad | m, n \geq 1, x \geq 0$

c> $a^n b^{n+2} \quad | n \geq 1$

Q.No.7

4 Marks

Write the Regular Expression for the following CFG and draw their DFA?

a> $S \rightarrow aSa$

$S \rightarrow \epsilon$

b> $S \rightarrow YXYXY$

$X \rightarrow a$

$Y \rightarrow bY \mid \epsilon$



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END SEMESTER EXAMINATIONS FALL 2021

Computer Science & Information
Technology

Faculty: FoCAS

Department

Semester 4th Degree Program BSCS

Batch 2020 S

Course Code: CS-226 Course Title: Software Engineering

Credit Hours: 3+0

Timings: 3 Hours

Date of Examination: 04-02-2022

Max Marks: 50

Instructions:

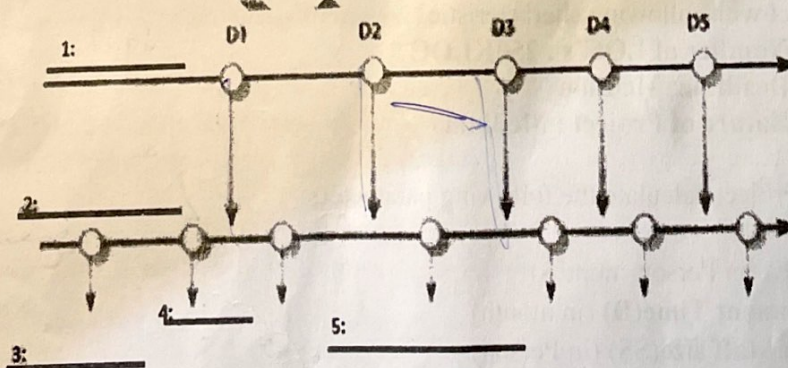
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Q.No.1

(05+08 marks)

- a) Software quality management is concerned with ensuring that software meets its required standards. What do you think "Required Standard" refers to? Explain with the help of some software applications.
- b) Complete the given process with appropriate terms and briefly explain each term.





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Q.No.2

(05+08 marks)

a) Some people argue that developers should not be involved in testing their own code but all testing should be the responsibility of a separate team. Give arguments for and against testing by the developers themselves.

b) Write possible test cases for the given interface of mobile banking.

The image shows a mobile banking login screen. It has a dark background with white text and icons. At the top, there is a 'Username' input field. Below it is a checked checkbox labeled 'Remember Username'. Then there is a 'Password (Alphanumeric)' input field with a lock icon on the left and an eye icon on the right. Below the password field is a section titled 'Select security image' with a question mark icon. This section contains a grid of ten icons: a lightbulb, a car, an airplane, a telephone, a bicycle, a helmet, glasses, a mug, an alarm clock, and a power button. Below the grid is a 'Log In' button. At the bottom, there is a link for 'Internet Banking Registration'. At the very bottom, there are three links: 'Forgot Username?', 'Forgot Password?', and 'Forgot Security Image?', followed by a 'Security Guide' link.

Q.No.3

(07 marks)

Consider Project with following characteristic

- Number of LOC's : 250KLOC
- Deadline: Medium
- Nature of Project : Medium

For the above Project calculate the following parameters:

- Effort(E) (in Person-months)
- Development Time(D) (in month)
- Average staff size(SS) (in Person)
- Productivity(P) (in KLOC/Person-month)



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Q.No.4

(07 marks)

Design a use case diagram for a centralized hotel reservation system which is working online. Make sure to mention at least 6 functionalities.

Q.No.5

(10 marks)

Given below are the feedbacks of different user of XYZ Component service provider. Read these feedbacks carefully and observe what basic characteristics these component must be providing. You can also find the missing elements/characteristics of components from these feedbacks. You are required to identify the component characteristics from given feedback with proper justification

- 1: I'm a one-person dev team, and I have another full-time job. I'm using XYZ Component Service Provider to support my website. My website has a lot of pages driven by data, and my website's feel-and-look has to be updated according to the constantly changing web trends. I like that XYZ Component service provider keep adding new features and new controls without delay to keep up with those trends.
- 2: This library is truly impressive with all its rich features. This comes with a variety of Controls which makes UI design so easier and effective. This enhances the look and feel of the application, led to high Customer satisfaction. I would also like to appreciate the Support team, who proactively provides complete solution with minimal time. It would be great if XYZ Component service provider will come with more specific documentation on generic problems
- 3: XYZ Component service provider's support is the best I've worked with. Their regular and valuable updates to the UI suite are also unmatched.
- 4 : An excellent tool with a broad and extensive documentation with decent support given the number of developers /businesses using the product. You can find a lot of documentation online as well. Also great integration with other SDK libraries.
- 5: Performance of the JavaScript framework for big data is an issue sometimes. Also, when using many components simultaneously slows down the rendering of the screen.



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Faculty: COMPUTER AND APPLIED SCIENCE Department: COMPUTER SCIENCE & IT

Semester: 4TH Degree Program: BS COMPUTER SCIENCE Batch: 2020S

Course Code: CS229 Course Title: DBMS Credit Hours: 3+1

Timings: 3 Hours Date of Examination: 16/02/2022 Max Marks: 50

Instructions:

- Please don't write anything on the Question Paper except for your Roll No. and Name.
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Roll No.	<u>214</u>	Name		'A' Copy Serial No.	
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Q.No.1 7 Marks

Consider the following tables, Rooms

(Room_id=primary key)

Room_id	Building	Room_number	Number_seats	Description
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Classes

Department	Course_description	Max_students	Current_students	Num_credits	Room_id
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Students

Roll_no	Student_name	Seat_No	Department
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Write the SQL commands for the following:

- Update classes with Current_students = 70 where Department = BM101
- Find the Room_number and the corresponding Building for the Course_description "DBMS"
Display the records of the table classes order by Department

Q.No.2 7 Marks

Develop the Entity Relation Diagram for the following scenario.

All relationship types of one to many, one to one and others required should be mentioned with all attributes, weak entities and details.

Consider a hospital:

Patients are treated in a single ward by the doctors assigned to them. Usually each patient will be assigned a single doctor, but in rare cases they will have two.



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The system must record details concerning patient treatment and staff payment. Some staff are paid part time and doctors and care assistants work varying amounts of overtime at varying rates (subject to grade).

The system will also need to track what treatments are required for which patients and when and it should be capable of calculating the cost of treatment per week for each patient (though it is currently unclear to what use this information will be put).

Q.No.3

7 Marks

Consider the relation $R \{A, B, C, D\}$ where A, B, C, D are attributes or fields. Following functional dependencies are given.

 $A \rightarrow BC$
 $B \rightarrow C$
 $A \rightarrow B$
 $AB \rightarrow C$
 $AC \rightarrow D$

a) Find out the irreducible set of functional dependencies.

b) Specify all the steps and rules to find out above irreducible set of functional dependencies.

Q.No.4

7 Marks

Scenario for normalization:

We have tables.

1. user_shirts (user_shirt_id (PK), user_id(FK), shirt_id(FK))
2. user_shirts_pants (user_shirts_pants_id (PK), user_shirt_id (Fk), pant_id(FK))
3. user_shirts_pants_belts (user_shirts_pants_belts_id(PK), user_shirts_pants_id (Fk), belt_id (Fk))

If there is

users(1) Shirts (A, B) pants (a,b,c) belts(I,II)

there will be 1 row in users table... there will be 2 rows in users_shirts table... there will be 6 rows in user_shirts_pants table... there will be 12 rows in user_shirts_pants_belts table....define attributes for shirt, pant and belt.

a) Specify the steps of 1st Normalization form, 2nd Normalization form, 3rd Normalization form and Boyce Codd Normalization form with reference to above scenario

b) What was the enhancement in 3rd Normalization form.

Q.No.5

7 Marks

When two transactions are accessing the records

- a) Show by timing diagrams valid timing for two transactions accessing the same account.
- b) Show the programming unit for the proper execution of transaction.
- c) Explain the ACID properties for the transactions and why they are important.

Q.No.6

7 Marks

Using JOINS, write down the QUERY/SQL command of the following output?

- a) -Write a query to display the employee last name, department name, location ID and city of all employees who earn a commission.

LAST NAME	DEPARTMENT NAME	LOCATION ID	CITY
Zlotkey	Sales	2500	Oxford
Abel	Sales	2500	Oxford
Taylor	Sales	2500	Oxford



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b) Write a query to display the last name, job, department number and department name for all employees who work in City Toronto.

LAST NAME	JOB ID	DEPARTMENT ID	DEPARTMENT NAME
Hartstein	MK_MAN	20	Marketing
Fay	MK_REP	20	Marketing

8 Marks

Q.No.7

Suppose we have 03 servers A, B, C by considering (current state) $X = PX^0$ in following case. Using stochastic model Predict the number of users/connections in server (by transition matrix) in a distributed database environment. Also calculate the number of users/connections at the end.

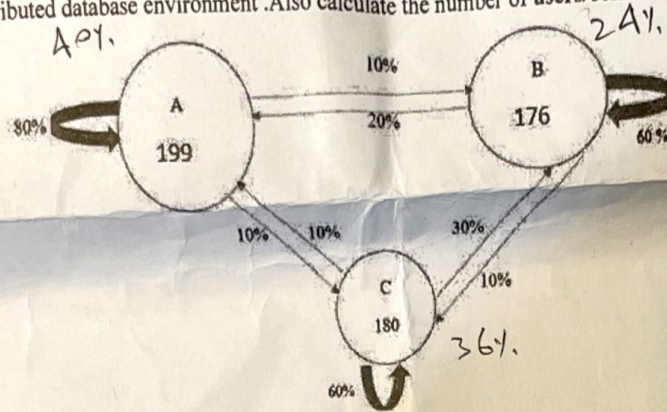


Figure 1